Graph Laplacian Regression for River-Network Water Quality Prediction

Colorado River Network: USGS Continuous Water-Quality Data + MERIT Hydro

Kennesaw State University – TDA Lab
Advisor: Dr. Mehmet Aktas

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Research Question

Goal: determine whether river-network structure improves prediction of water-quality variables beyond ordinary regression on contemporaneous chemistry.

Two prediction settings emerged:

Sensor-assisted prediction

Predict one water-quality variable when the other four are observed at the same hour.

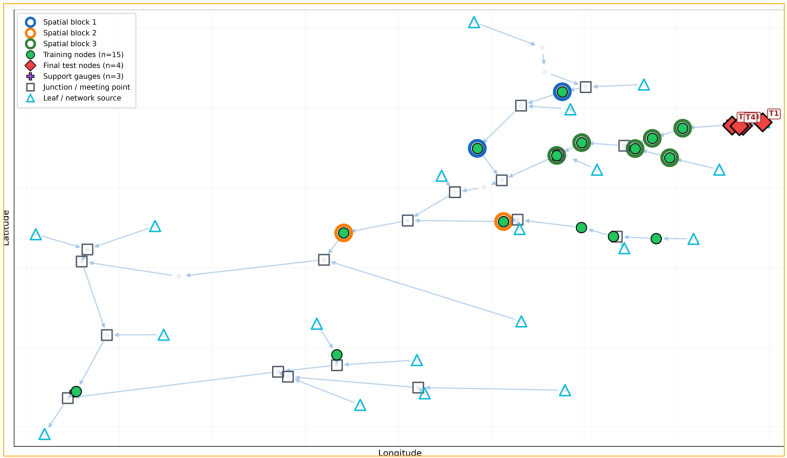
Spatial/network contribution

Compare ordinary Ridge regression against a model with a graph-Laplacian spatial effect under held-out river regions.

Main targets:

$$\text{pH},\ \text{DO},\ \text{Water Temperature},\ \text{Specific Conductance},\ \text{Turbidity}.$$

River Network and Spatial Validation Design



67-node compressed graph. Colored rings indicate spatial CV blocks; green circles are development/training nodes, red diamonds are the four final holdout nodes, purple markers are hydrology-support gauges, gray squares are junctions, and cyan triangles are network sources/endpoints.

Hourly Water-Quality Dataset

- 127,454 hourly rows contain pH.
- 84,855 rows contain pH plus all four chemistry predictors.
- Complete-case pH data span 16 USGS stations.
- After station-to-node aggregation and spatial-development filtering:

  52,949 node-hours across 10 graph nodes.

- Spatial blocks:
  - Block 1: 2 nodes, 13,518 node-hours
  - Block 2: 2 nodes, 10,932 node-hours
  - Block 3: 6 nodes, 28,499 node-hours

Hourly predictors

For each target, the other four water-quality variables are used as predictors.

$$X_{i,t} = \begin{bmatrix} x_{1,i,t} & x_{2,i,t} & x_{3,i,t} & x_{4,i,t} \end{bmatrix}$$

Models

Linear Ridge regression

$$y_{i,t} = X_{i,t}\beta + \varepsilon_{i,t}.$$

Linear + Graph Laplacian

$$y_{i,t} = X_{i,t}\beta + g_i + \varepsilon_{i,t},$$

$$\sum_i \frac{1}{n_i} \sum_t (y_{i,t} - X_{i,t}\beta - g_i)^2 + \lambda_L g^\top L g + \lambda_\beta \|\beta\|_2^2, \qquad \sum_i g_i = 0.$$

- L: distance-weighted graph Laplacian.
- Edge similarity: $w_{ij} = \exp(-d_{ij}/\tau)$.
- Equal total weight per graph node prevents heavily sampled stations from dominating.

Spatial Validation

Leave-one-spatial-block-out CV

For each fold:

1. hold out an entire river-network block,
2. fit on the remaining blocks,
3. predict every hourly observation in the held-out block,
4. weight validation so each graph node contributes equally.

Why this matters

Random hourly splitting would allow observations from the same monitoring locations into both train and test sets.

Spatial block CV instead asks:

Does the relationship transfer to an unseen river region?

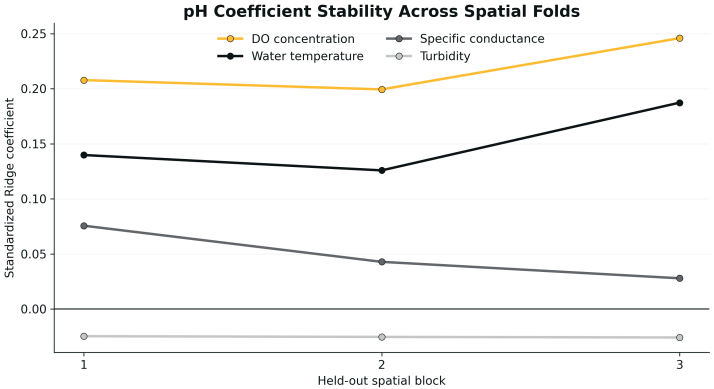
pH Model Search

Model	RMSE	MAE	R^2
Linear chemistry	**0.1584**	**0.1293**	**0.438**
Linear + Laplacian	0.1587	0.1296	0.436
Seasonal linear model	0.1714	0.1380	0.342
Gradient boosting	0.1897	0.1548	0.194
Quadratic Ridge	0.2239	0.1864	-0.123

Takeaway

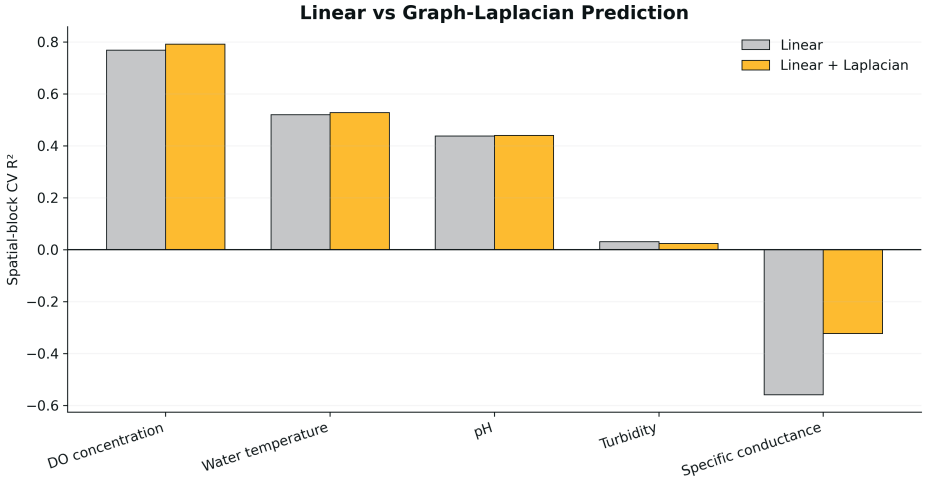
For hourly pH prediction, contemporaneous chemistry contains most of the useful signal. More complex temporal, nonlinear, and graph-regularized models did not improve spatial generalization.

pH Coefficients Are Spatially Stable

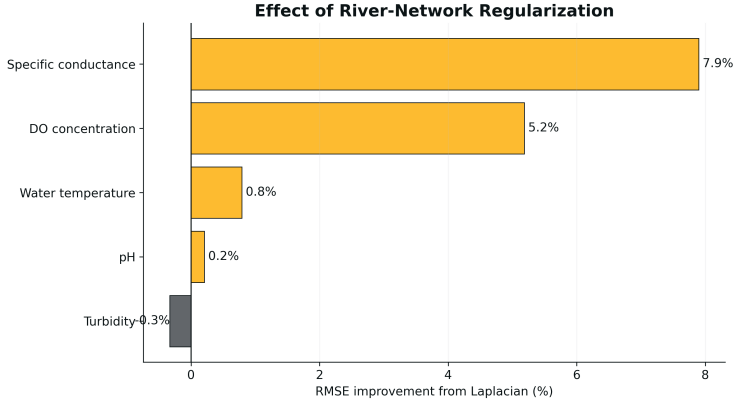


All four standardized coefficients preserved their sign across the three held-out spatial folds. Dissolved oxygen had the largest positive conditional association with pH, followed by water temperature.

Does the Graph Help Across Water-Quality Targets?



RMSE Effect of Graph-Laplacian Regularization



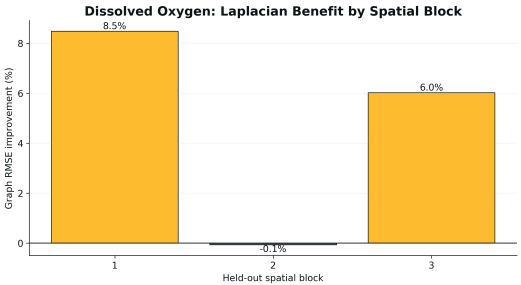
Positive values mean the Laplacian reduced RMSE. Dissolved oxygen showed the clearest useful improvement among well-performing targets.

Target-by-Target Results

Target	Linear R^2	Graph R^2	Linear RMSE	Graph RMSE
DO concentration	0.769	**0.792**	0.835	**0.791**
Water temperature	0.520	**0.528**	4.521	**4.485**
pH	0.438	**0.440**	0.1584	**0.1581**
Specific conductance	-0.559	-0.322	380.99	**350.90**
Turbidity	**0.031**	0.024	**344.82**	345.96

- DO: meaningful improvement with graph structure.
- Temperature and pH: only marginal graph benefit.
- Conductance: graph reduces error substantially, but absolute generalization remains poor.
- Turbidity: graph does not improve prediction.

Strongest Graph Result: Dissolved Oxygen



Best graph model:

$$\lambda_L = 10, \qquad \tau = 50 \text{ km}, \qquad \lambda_\beta = 1.$$

Overall:

$$R^2 : \ 0.769 \rightarrow 0.792$$

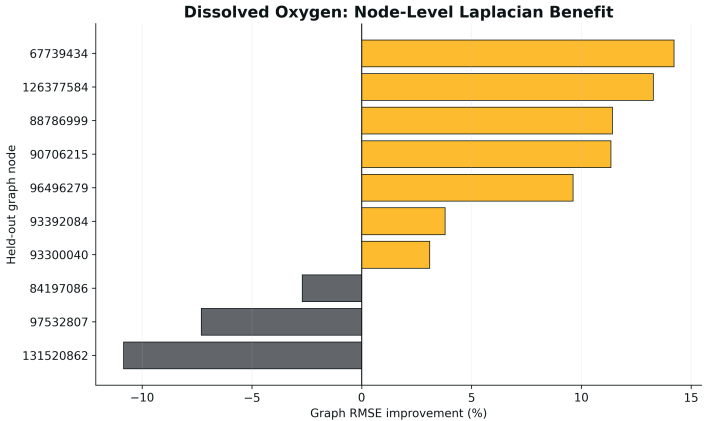
$$\mathrm{RMSE} : \ 0.835 \rightarrow 0.791$$

5.19% lower RMSE

Block-level RMSE improvement:

$$8.49\%, \quad -0.07\%, \quad 6.02\%.$$

DO Improvement Is Not Driven by One Station



The Laplacian improved DO prediction at 7 of 10 held-out graph nodes. Median node-level RMSE improvement was approximately 6.7%.

What Explains the DO Improvement?

We compared graph benefit against:

- upstream drainage area,
- elevation,
- river width,
- source/outlet distance,
- nearest training-node distance,
- upstream/downstream training coverage.

Result

No simple static or proximity variable clearly explained which nodes benefited from the Laplacian.

All held-out nodes were far from training observations (roughly 370–704 km to the nearest training node), and none had a training node within 150 km.

Interpretation: the DO graph benefit is real overall, but the current 10-node spatial validation sample is too small to reliably identify which site characteristics determine that benefit.

Static River Features and Fully Unmonitored Prediction

For pH, we also tested predictors available everywhere on the river graph:

$$\log(1 + \mathrm{UPA}), \quad \log(1 + \mathrm{width}), \quad \text{elevation}, \quad \text{network position}.$$

Best deployable static model:

$$\log(1 + \mathrm{UPA}) + \log(1 + \mathrm{width}) + \text{Laplacian}.$$

Spatial-block result:

$$\mathrm{RMSE} = 0.1366, \qquad R^2 = -0.084.$$

Takeaway

Static river features can be evaluated at completely unmonitored nodes, but they did not explain regional pH variation as well as contemporaneous chemistry.

Main Conclusions

1. **Graph value is target-dependent.**
2. **Dissolved oxygen is the strongest case for graph-Laplacian regularization:**

   5.2% lower RMSE overall, improvement at 7/10 nodes.

3. For hourly pH, same-hour chemistry is much more important than graph structure.
4. More complex nonlinear or seasonal models did not improve pH spatial generalization.
5. Spatial-block validation reveals weaknesses that random hourly splits would hide.

Limitations and Next Steps

- Current complete-case spatial experiment uses only 10 development graph nodes.
- Chemistry-assisted models require the other water-quality variables at the prediction site.
- Conductance and turbidity remain difficult to generalize spatially.
- Temporal coverage differs across monitoring locations.

Next steps:

- expand the number of stations and target-specific usable hours,
- evaluate flow-directed or hydrology-weighted Laplacians,
- reserve a fresh external spatial test set,
- investigate multivariate / multi-output graph models.

Thank You

Questions?